Field Hours

CSCE 640 Quantum Algorithms

Original notes for this site. Same topics as the catalog course. Not a reprint of another university's lectures, and not academic credit.

Lecture 1 · 90 minutes

One qubit, one gate X

You write a one-qubit state as a|0> + b|1>, with real amplitudes whose squares sum to 1. The state |0> has a = 1 and b = 0. A circuit is an ordered list of gates. The gate X swaps the two amplitudes, so X applied to |0> returns |1>. Measurement returns 0 with probability a squared and 1 with probability b squared. After one X those probabilities are 0 and 1.

You apply gates from the input outward, one at a time, on a state you can write. On this circuit the input is |0>, the list is a single X, and the output is |1>. Measurement returns the bit 1 with probability 1. A second X swaps back to |0>, so the probability of 0 is again 1. Two gates on one qubit give you an identity you can check without any hidden notation.

Send |0> through X

Start in state |0> -> Apply the gate X -> State is now |1> -> Measure the bit 1

X, then X again

1. Start at |0>: amplitude 1 on |0> and amplitude 0 on |1>.

2. One X swaps them. A measurement returns 1 with probability 1 squared, which is 1.

3. A second X swaps back to |0>. Two X gates cancel, and the probability of measuring 0 is 1.

Problem 1. You apply X three times to |0>. What is the state, and what is the probability of measuring 1?

Answer. Three swaps end at |1|. The probability of measuring 1 is 1. Two swaps would have returned |0>.

Problem 2. A state has amplitude 0 on |0> and amplitude -1 on |1>. What is the probability of measuring 1?

Answer. (-1) squared is 1. The sign is invisible to the probability. You measure 1 with certainty.

Lecture 2 · 90 minutes

Hadamard, and a second Hadamard

You use the Hadamard gate H to send |0> to (|0> + |1>) divided by the square root of 2, and to send |1> to (|0> - |1>) divided by the same square root. Each outcome then has probability 1/2, because each amplitude squares to 1/2. H builds a uniform superposition. It does not itself emit a random bit. The measurement is what turns the two amplitudes into a bit.

You cancel two Hadamard gates. H on |0> produces the superposition above. H on that result sends the |0> piece and the |1> piece through H again. The |1> amplitudes cancel, and the |0> amplitudes add to 1, so the state returns to |0>. Measurement then returns 0 with probability 1. The circuit of 2 Hadamards is the identity on this input. You checked it by adding two explicit states.

One H, then another

After one H

Each outcome has chance 1/2

State is not |0>

After two H gates

Back to state |0>

Measure 0 for certain

H twice returns |0>

1. Start at |0> and apply H. The state is (|0> + |1>) divided by the square root of 2.

2. Apply H again. The |1> amplitudes cancel and the |0> amplitudes add to 1.

3. The state is |0>. Measurement returns 0 with probability 1. You used 2 gates.

Problem 1. Apply H once to |0> and measure. What is the probability of outcome 1?

Answer. The amplitude of |1> is 1 over the square root of 2. Its square is 1/2.

Problem 2. The input is |0> and you want |0> at the output using only H. How many H gates do you use, and what is the probability of measuring 0?

Answer. Two. One H leaves the superposition. The second returns |0>, and the probability of measuring 0 is 1.

Lecture 3 · 90 minutes

Deutsch on a one-bit function

You use Deutsch's algorithm to test a function f from one bit to one bit. Constant means f(0) equals f(1). Balanced means the two values differ. Classically the worst case takes 2 evaluations. The quantum circuit uses 1 query to an oracle that sends |x> |y> to |x> |y xor f(x)>. You put the target in (|0> - |1>) over the square root of 2. The oracle then multiplies |x> by -1 to the power f(x).

You put H on the input, query once, and put H on the input again. If f is constant 0, every phase is +1, the two H gates cancel, and you measure 0. Constant 1 multiplies the whole state by -1, a sign you cannot see, and you still measure 0. If f(x) = x, the oracle negates only |1>, and the final H yields |1>. You measure 1 and report balanced. One query names the type.

One query, then measure

1. H on the input qubit

2. Query the oracle once

3. H on the input again

4. 0 constant, 1 balanced

The balanced function f(x) = x

1. f(0) = 0 and f(1) = 1, so f is balanced. After the first H the input is (|0> + |1>) over the square root of 2.

2. The oracle multiplies the |1> term by -1. The input becomes (|0> - |1>) over the square root of 2.

3. The final H turns that state into |1>. You measure 1 and report balanced, after 1 query.

Problem 1. f(0) = 1 and f(1) = 1. What do you measure on the input qubit, and which type is f?

Answer. You measure 0. The values match, so f is constant. The shared minus sign is global and does not change the measurement.

Problem 2. Why do 2 classical queries decide the type, and why is 1 classical query not enough in the worst case?

Answer. Two queries read f(0) and f(1). Matching values mean constant, and differing values mean balanced. After one classical query both types remain possible.

Lecture 4 · 90 minutes

The same promise on two bits

You extend the promise to n = 2 bits, where a function has 4 inputs. Constant means one output on all 4. Balanced means the output 0 appears on exactly 2 inputs. Deutsch-Jozsa uses 1 quantum query. You measure the all-zero string when f is constant, and some nonzero string when f is balanced. You are separating a promise, not identifying an arbitrary function of 4 inputs.

You need 3 classical queries in the worst case on these 4 inputs. Two answers of 0 still leave both promises open: the hidden pair might be 0, 0, which is constant, or 1, 1, which is balanced. A third query breaks the tie. Under the promise you never need the fourth query, so the classical number is 3 against 1 quantum query. Drop the promise and one quantum query no longer names an arbitrary f.

Two zeros, both promises

Could be constant

Seen outputs 0 and 0

Hidden pair is 0, 0

Could be balanced

Seen outputs 0 and 0

Hidden pair is 1, 1

Why the third query exists

1. You query two of the four inputs and both return 0.

2. Constant 0 still fits if the hidden pair is 0, 0. Balanced still fits if the hidden pair is 1, 1.

3. A third classical query breaks the tie. The quantum circuit still uses 1 query.

Problem 1. You have seen outputs 0, 0, and 1 on three inputs, and the promise holds. Is f constant or balanced, and do you query the fourth input?

Answer. Balanced. Constant is already impossible because you saw both 0 and 1. The unseen input must be 1 so that exactly two inputs return 1. You do not need to query it.

Problem 2. How many quantum queries does Deutsch-Jozsa use for n = 2, and how many classical queries are in the worst case?

Answer. 1 quantum query, and 3 classical queries. All-zero measurement means the promised function is constant.

Lecture 5 · 90 minutes

Grover on four marked-or-not items

You search N = 4 items for one marked item with Grover's iterate. Start from the uniform superposition, amplitude 1/2 on each item. The oracle flips the sign of the marked item. Diffusion replaces each amplitude a by 2 m - a, where m is the mean of the four amplitudes. One query finishes this N. Classically, with exactly one marked item, the worst case still looks at 3 items before the last item is forced.

You compute the amplitudes. After the sign flip they are 1/2, 1/2, 1/2, and -1/2. The sum is 1 and the mean m is 1/4. An unmarked amplitude becomes 2 times 1/4 minus 1/2, which is 0. The marked amplitude becomes 2 times 1/4 minus -1/2, which is 1. You measure the marked item with probability 1. The other 3 amplitudes are 0. For larger N you would repeat about square root of N times. Here 1 iterate is exact.

One iterate on four items

Start at amplitude 1/2 -> Flip the marked sign -> Reflect about the mean -> Measure the marked item

Mean one fourth, then a spike

1. After the oracle the amplitudes are 1/2, 1/2, 1/2, -1/2. Sum 1, mean 1/4.

2. Unmarked update: 2 times 1/4 minus 1/2 = 0. Marked update: 2 times 1/4 plus 1/2 = 1.

3. Measurement returns the marked item with probability 1. You used 1 oracle query.

Problem 1. Before diffusion, with amplitudes 1/2, 1/2, 1/2, -1/2, what is the probability of measuring the marked item?

Answer. (-1/2) squared is 1/4. Each unmarked item also has probability 1/4. The sign flip alone does not concentrate the probability.

Problem 2. After diffusion, how many of the 4 amplitudes are 0, and what is the probability of a wrong item?

Answer. The other 3 amplitudes are 0. The probability of a wrong item is 0. The marked probability is

1.

Lecture 6 · 90 minutes

A Bell state is not two copies

You treat (|00> + |11>) over the square root of 2 as one joint state. The probability of 00 is 1/2 and of 11 is 1/2. The probability of 01 or of 10 is 0. Measure the first qubit as 0 and the second is |0>. Measure 1 and the second is |1>. Two independent copies of (|0> + |1>) over the square root of 2 would also yield 01 and 10. This state does not.

You cannot clone an unknown qubit with a linear circuit. Suppose a map sends |0>|0> to |0>|0> and |1>|0> to |1>|1>. On (|0> + |1>) over the square root of 2, times |0>, linearity emits (|00> + |11>) over the square root of 2. A real copy would be the product of the superposition with itself, four basis terms each of amplitude 1/2. Two terms are not four terms. The map made the Bell state, not two copies.

Four terms versus two

A real copy

Four basis terms

Each amplitude is 1/2

The linear image

Only |00> and |11>

01 and 10 have chance 0

Why the cloner misses |+>

1. The input is (|0> + |1>) over the square root of 2, beside a fresh |0>.

2. Linearity emits (|00> + |11>) over the square root of 2.

3. A true copy expands to four basis states, each of amplitude 1/2. Two terms are not that copy.

Problem 1. You measure (|00> + |11>) over the square root of 2 and the first qubit is 1. What is the second qubit, and what is the probability of this branch?

Answer. The second qubit is |1>. The probability is 1/2, the square of 1 over the square root of 2. Outcome 10 has probability 0.

Problem 2. How many basis terms are in the product of the superposition with itself, and how many are in the Bell state?

Answer. The product has 4 terms. The Bell state has 2. The linear map lands on the 2-term state, not on the 4-term copy.

Lecture 7 · 90 minutes

H then CNOT builds the pair

You prepare that Bell state with 2 gates from |00>. Hadamard on the first qubit yields (|00> + |10>) over the square root of 2. CNOT, control on the first qubit and target on the second, turns |10> into |11> and leaves |00> alone. The state is (|00> + |11>) over the square root of 2. H made the superposition. CNOT wrote the control bit onto the target.

You still measure 00 or 11, each with probability 1/2. This circuit uses 0 oracle queries, 1 Hadamard, and 1 CNOT. If the first qubit measures 0, the second is |0> on a branch of probability 1/2. Order matters: CNOT first on |00> does nothing, and a later H leaves (|00> + |10>) over the square root of 2. That is a product state, not the Bell state.

Two gates from |00>

Start from |00> -> H on the first qubit -> CNOT onto the second -> Two nonzero amplitudes

Build, then read 00

1. After H, |00> and |10> each have amplitude 1 over the square root of 2.

2. CNOT sends |10> to |11> and leaves |00> fixed.

3. The probability of 00 is 1/2 and the probability of 11 is 1/2. Amplitude on |01> and on |10> is 0.

Problem 1. What is the probability of measuring 01 after H then CNOT on |00>?

Answer. 0. The only nonzero amplitudes are on |00> and |11>, each with probability 1/2.

Problem 2. You apply CNOT before H, starting from |00>, same control and target. What state do you get?

Answer. CNOT leaves |00> unchanged. H then yields (|00> + |10>) over the square root of 2. That is a product state, not the Bell state. The order was the mechanism.

Lecture 8 · 90 minutes

What you can now calculate

You can now apply X and H, read a probability as an amplitude squared, and run Deutsch's one-query test. You can run one Grover iterate on N = 4, where the marked amplitude becomes 1, and you can build (|00> + |11>) over the square root of 2 with H then CNOT. A probability is a square: 1 over the square root of 2, squared, is 1/2. Every amplitude in these circuits was written down.

You add query bills without mixing promises. Deutsch uses 1 quantum query against a classical worst case of 2. Grover on 4 items uses 1 quantum query against a classical worst case of 3. The quantum total is 2 and the classical worst-case total is 5. The sum is not a new algorithm. One promise is constant versus balanced. The other is one marked item among 4.

Add the two query bills

Quantum queries

Deutsch costs 1

Grover costs 1

Classical worst case

Deutsch costs 2

Grover costs 3

Two queries against five

1. Deutsch queries once and measures 0 on a constant function.

2. Grover queries once on N = 4 and leaves the marked amplitude at 1.

3. Quantum total: 1 + 1 = 2. Classical worst-case total: 2 + 3 = 5. The gap on this pair is 3 queries.

Problem 1. You run Deutsch and the 4-item Grover search as above. How many quantum queries, and how many classical worst-case queries, do you spend in total?

Answer. Quantum: 1 + 1 = 2. Classical worst case: 2 + 3 = 5.

Problem 2. Combine the Bell state with Deutsch's outcome. The probability of measuring 11 on (|00> + |11>) over the square root of 2 is what, and why is that not the same fact as Deutsch measuring 1 on a balanced function?

Answer. The Bell probability is 1/2. Deutsch's 1 is a one-qubit label for a balanced function, not a probability of 1/2 and not a two-qubit outcome. The shared symbol 1 does not make them one experiment.